International Islamic University Chittagong

Department of Computer Science & Engineering

Mid Term Examination, Autumn 2025

CSE 2422 Computer Algorithms Lab

Total marks: **30** Time: **60** minutes

[Answer all of the following questions. Figures in the right-hand margin indicate full marks.]

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1. Consider a imaginary Fibonacci number sequence whose recurrence relation is CLO2

$$f(n) = \begin{cases} n, & \text{if } n \leq 2 \\ f(n-1) + f(n-2) + f(n-3), & \text{if } n > 2 \end{cases}$$

Now, find the 50th Fibonacci number using this recursive equation.

2. Given an array of n elements, implement insertion sort and selection sort algorithm and compare their runtime. CLO2
Or
Implement any two sorting algorithms.

3. You are given strings *ss* and *tt*. Find one longest string that is a subsequence of both *ss* and *tt*. Complete the following code below: CLO3

Input copy	Output copy
axyb abyxb	axb